HDOC Day-42 - C Programming

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Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Remove all vowels from a string.
2. Find the first repeating lowercase alphabet in a string.
3. Check if two strings are anagrams of each other.
4. Exit.

Header Files Used: stdio.h, string.h, ctype.h

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void removeVowels(char str[])
{
    int i, j = 0;
    char temp[100];

    for (i = 0; str[i] != '\0'; i++)
    {
        char ch = tolower(str[i]);

        if (ch != 'a' && ch != 'e' && ch != 'i' &&
            ch != 'o' && ch != 'u')
        {
            temp[j] = str[i];
            j++;
        }
    }

    temp[j] = '\0';
    strcpy(str, temp);
}

char firstRepeating(char str[])
{
    int count[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            count[str[i] - 'a']++;

            if (count[str[i] - 'a'] == 2)
                return str[i];
        }
    }

    return '\0';
}

int checkAnagram(char str1[], char str2[])
{
    int count[26] = {0};
    int i;

    for (i = 0; str1[i] != '\0'; i++)
    {
        if (isalpha((unsigned char)str1[i]))
            count[tolower((unsigned char)str1[i]) - 'a']++;
```

```
    }

    for (i = 0; str2[i] != '\0'; i++)
    {
        if (isalpha((unsigned char)str2[i]))
            count[tolower((unsigned char)str2[i]) - 'a']--;
    }

    for (i = 0; i < 26; i++)
    {
        if (count[i] != 0)
            return 0;
    }

    return 1;
}

int main()
{
    char str1[100], str2[100];
    char ch;
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find first repeating lowercase alphabet\n");
        printf("3. Check two strings are anagrams\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                removeVowels(str1);
                printf("String after removing vowels: %s\n", str1);
                break;

            case 2:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                ch = firstRepeating(str1);

                if (ch != '\0')
                    printf("First repeating lowercase alphabet: %c\n", ch);
                else
                    printf("No repeating lowercase alphabet found.\n");
                break;

            case 3:
                printf("Enter first string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                printf("Enter second string: ");
                fgets(str2, sizeof(str2), stdin);
                str2[strcspn(str2, "\n")] = '\0';

                if (checkAnagram(str1, str2))
                    printf("The strings are anagrams.\n");
                else
                    printf("The strings are not anagrams.\n");
                break;

            case 4:
                printf("Program exited successfully.\n");
                break;
            default:
```

```
                printf("Invalid choice.\n");
        }

    } while (choice != 4);

    return 0;
}
```

Program Output

